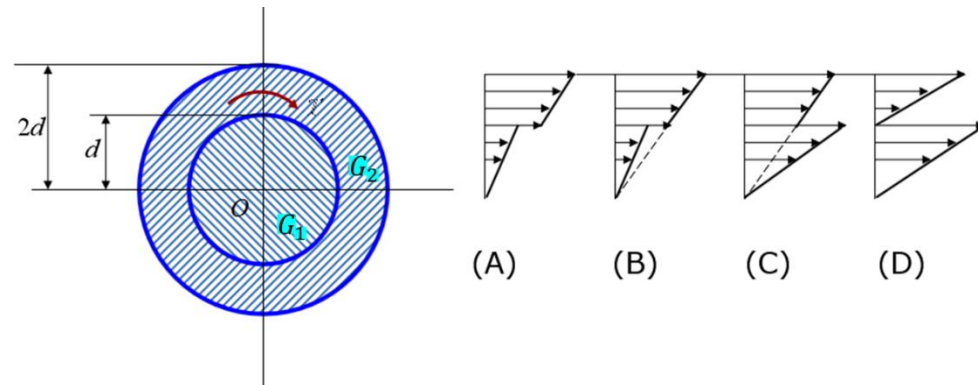


Submission: Please submit the scanned/typed document to Canvas.

1. A tube made of two different materials. The shear modulus of the outer and inner one are G_1 and G_2 , respectively. And $G_1 = 2G_2$. When a torsional loading is applied, there is no relative sliding. For the shear stress distribution on the cross-section, which one is correct?



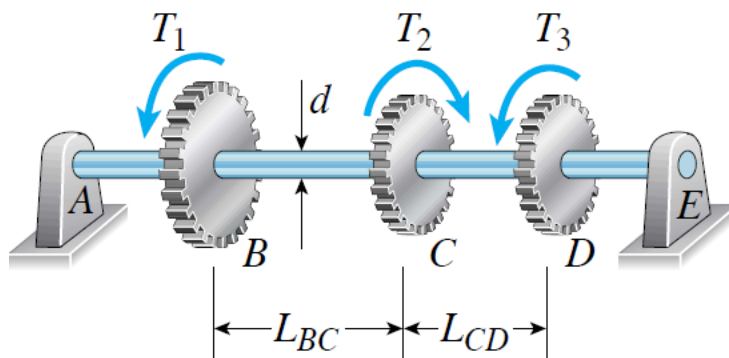
Correct Answer: C

When the circular shaft is under torsion, there is no relative sliding between the inner and outer layers. This means they deform together as a single body, experiencing torsional deformation simultaneously. According to the plane section assumption, the composite cross-section remains plane after torsional loading. That is, radial lines remain straight lines, but each line rotates through some angle from its original position.

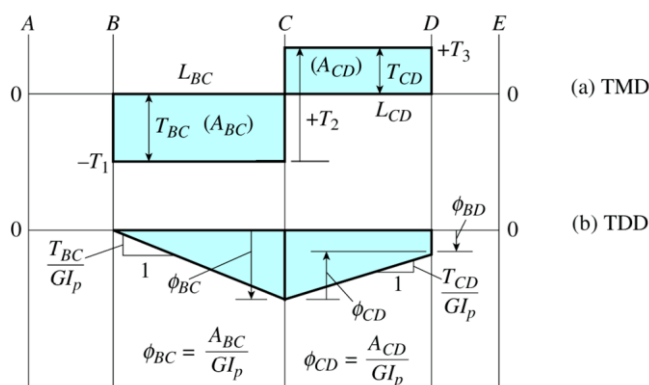
Therefore, at the interface between the inner and outer layers, both materials must exhibit the same shear strain. Since the shear modulus of the inner material (the solid core) is greater than that of the outer material ($G_1 = 2G_2$), the shear stress in the inner layer at the interface must be larger than that in the outer layer. Thus, option (A) and option (B) are incorrect because they do not satisfy this condition. In option (D), the shear stress in the outer layer at the interface is shown as zero, which is also incorrect, since the shear strain continuity requires nonzero shear stress there. By the shear stress–strain law, the shear stress cannot be zero at the interface.

2. Figure shows a solid steel transmission shaft with a diameter $d = 30$ mm. The driving gear is C, passive gears are B and D. $T_1 = 275$ N·m, $T_2 = 450$ N·m, $T_3 = 175$ N·m. $L_{BC} = 500$ mm, $L_{CD} = 400$ mm. Shear modulus $G = 80$ GPa. Determine

- the $\tau_{\max}$ in each part of the shaft
- angle of twist between gears B and D



Solution:



We have

$$\tau_{\max} = \frac{Tr}{I_p} = \frac{T \frac{d}{2}}{\frac{\pi}{32} d^4} = \frac{16T}{\pi d^3}$$

Therefore, the magnitude of maximum shear stress

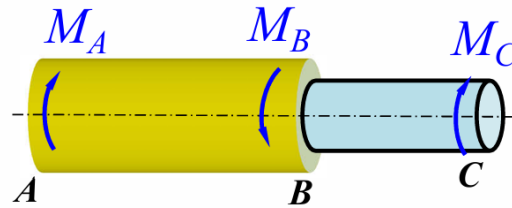
$$\tau_{\max,BC} = \frac{16|T_{BC}|}{\pi d^3} = \frac{16T_1}{\pi d^3} = \frac{16(275 \text{ N} \cdot \text{m})}{\pi(30 \text{ mm})^3} = 51.9 \text{ MPa}$$

$$\tau_{\max,CD} = \frac{16|T_{BC}|}{\pi d^3} = \frac{16T_3}{\pi d^3} = \frac{16(175 \text{ N} \cdot \text{m})}{\pi(30 \text{ mm})^3} = 33.0 \text{ MPa}$$

The angle of twist ϕ_{BD}

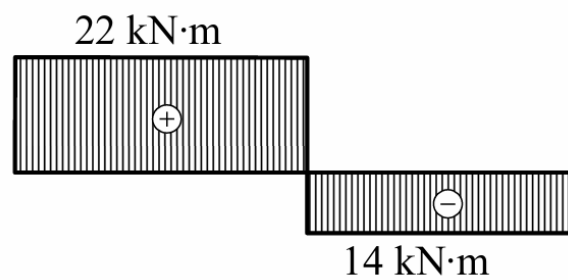
$$\begin{aligned} \phi_{BD} &= \phi_{BC} + \phi_{CD} = \frac{T_{BC}L_{BC}}{GI_p} + \frac{T_{CD}L_{CD}}{GI_p} = \frac{-T_1L_{BC}}{GI_p} + \frac{T_3L_{CD}}{GI_p} = -0.0216 + 0.0110 \text{ rad} \\ &= -0.0106 \text{ rad} = -0.61^\circ \end{aligned}$$

3. Given: AB segment diameter $d_1 = 120$ mm, BC segment diameter $d_2 = 100$ mm. $M_A = 22$ kN·m, $M_B = 36$ kN·m, $M_C = 14$ kN·m. The allowable stress $[\tau] = 80$ MPa. Determine if the structure is safe.



Solution:

TMD:



The magnitude of maximum shear stress

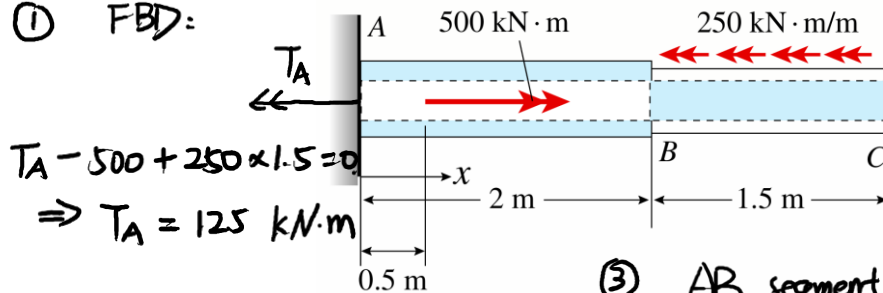
$$\tau_{\max,AB} = \frac{16|T_{AB}|}{\pi d_1^3} = \frac{16(22 \times 10^3 \text{ N} \cdot \text{m})}{\pi(120 \text{ mm})^3} = 64.84 \text{ MPa} < [\tau] = 80 \text{ MPa}$$

$$\tau_{\max,BC} = \frac{16|T_{BC}|}{\pi d_2^3} = \frac{16(14 \times 10^3 \text{ N} \cdot \text{m})}{\pi(100 \text{ mm})^3} = 71.3 \text{ MPa} < [\tau] = 80 \text{ MPa}$$

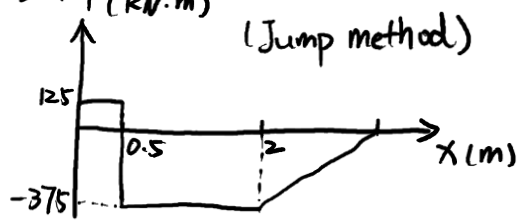
So the structure is safe.

4. Two sections of steel drill pipe, joined by bolted flange plates at B, are being tested to assess the adequacy of both the pipes. In the test, the pipe structure is fixed at A, a concentrated torque of $500 \text{ kN}\cdot\text{m}$ is applied at $x = 0.5 \text{ m}$, and uniformly distributed torque intensity $t_0 = 250 \text{ kN}\cdot\text{m/m}$ is applied on pipe BC. Both pipes have the same inner diameter $d = 200 \text{ mm}$. Pipe AB has thickness $t_{AB} = 15 \text{ mm}$, while pipe BC has thickness $t_{BC} = 12 \text{ mm}$. Find the maximum shear stress and maximum twist of the pipe and their locations along the pipe. Assume $G = 75 \text{ GPa}$.

① FBD:

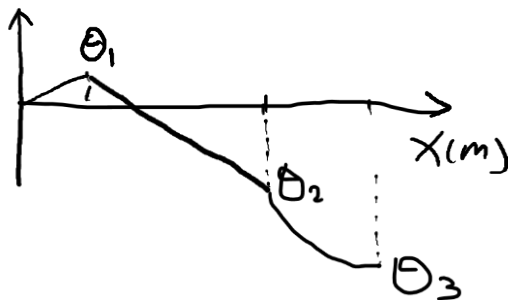


② TMD: $T (\text{kN}\cdot\text{m})$



④ $\theta = \int_0^x \frac{T(x)}{GI_p} dx$

TMD:



$$\theta_1 = \frac{(125 \text{ kN}\cdot\text{m}) \times 0.5 \text{ m}}{GI_p} = 7.06 \times 10^{-3} \text{ rad}$$

$$\theta_2 = \theta_1 + \frac{(-375 \text{ kN}\cdot\text{m}) \times 1.5 \text{ m}}{GI_p} = -5.65 \times 10^{-2} \text{ rad}$$

$$\theta_3 = \theta_2 + \int_2^{3.5} \frac{1}{GI_p} [250 \text{ kN}\cdot\text{m/m} \cdot (x - 3.5)] dx$$

$$= \theta_2 - 4.46 \times 10^{-2} \text{ rad} = 0.10 \text{ rad}$$

③ AB segment:

$$|T|_{\max} = \frac{|T|_{\max} r_{AB}}{I_{p,AB}}$$

$$r = \frac{d}{2} + t_{AB} = 0.115 \text{ m}$$

$$I_p = \frac{\pi}{32} (0.23^4 - 0.2^4) \approx 1.18 \times 10^{-4} \text{ m}^4$$

$$|T|_{\max} = 375 \text{ kN}\cdot\text{m}$$

$$\Rightarrow |T|_{\max} = 365.5 \text{ MPa}$$

BC segment:

$$|T|_{\max} = \frac{|T|_{\max} r_{BC}}{I_{p,BC}}, \quad |T|_{\max} = 375 \text{ kN}\cdot\text{m}$$

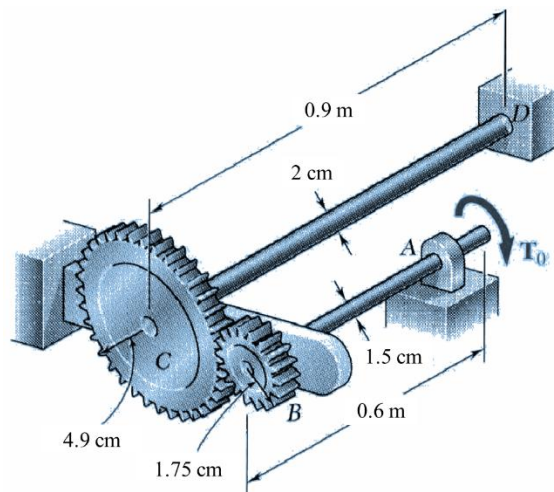
$$r_{BC} = \frac{d}{2} + t_{BC} = 0.112 \text{ m}$$

$$I_p = \frac{\pi}{32} (0.224^4 - 0.2^4) = 9.01 \times 10^{-5} \text{ m}^4$$

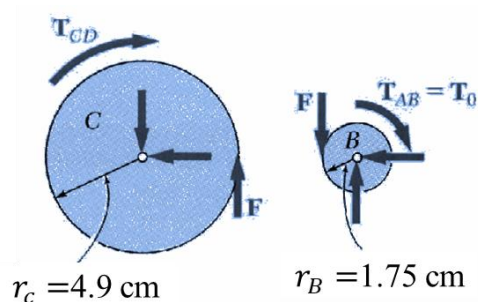
$$\Rightarrow |T|_{\max} = 466.2 \text{ MPa}$$

5. Two solid steel shafts are connected by gears. Knowing that for each shaft $G = 75 \text{ GPa}$, and the allowable shear stress $[\tau] = 100 \text{ MPa}$. Other geometric information is given in figure. Determine

- The largest torque T_0 that may be applied to the end of shaft AB
- The corresponding angle of twist ϕ_A at end A.



Solution:



Static equilibrium for two shafts:

$$\sum M_B = 0 = Fr_B - T_0$$

$$\sum M_C = 0 = Fr_C - T_{CD}$$

$$\Rightarrow T_{CD} = 2.8T_0$$

Also, from geometry

$$r_B \phi_B = r_C \phi_C \Rightarrow \phi_B = 2.8\phi_C$$

For AB segment, $T_{AB} = T_0$,

$$\tau_{max} = \frac{T_{AB} r_{AB}}{I_{p,AB}} = \frac{T_0 (0.0075 \text{ m})}{\frac{\pi}{32} (0.015)^4} = [\tau] \Rightarrow T_0 = 66.27 \text{ N} \cdot \text{m}$$

For CD segment,

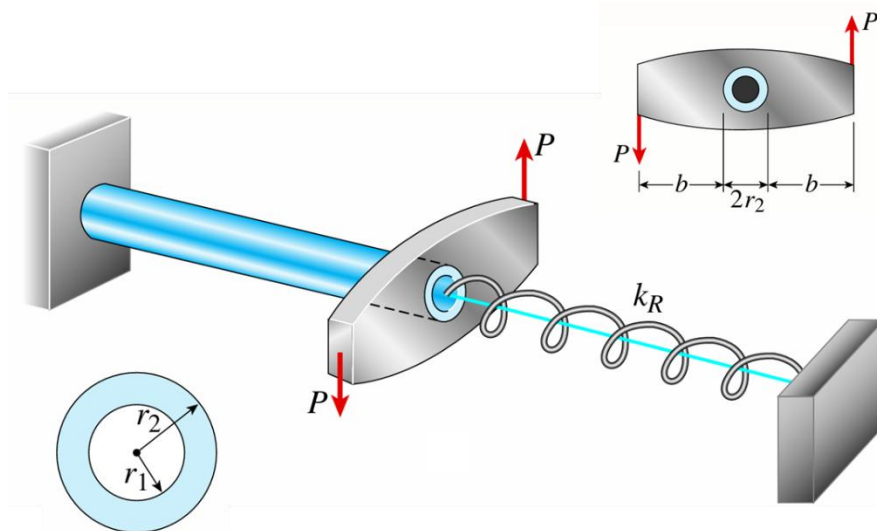
$$\tau_{max} = \frac{T_{CD} r_{CD}}{I_{p,CD}} = \frac{2.8T_0 (0.01 \text{ m})}{\frac{\pi}{32} (0.02)^4} = [\tau] \Rightarrow T_0 = 56.10 \text{ N} \cdot \text{m}$$

Therefore the largest torque $T_0 = 56.10 \text{ N} \cdot \text{m}$.

$$\phi_C = \frac{T_{CD} L_{CD}}{G I_{p,CD}} = \frac{2.8 \times 56.10 \text{ N} \cdot \text{m} \times 0.9 \text{ m}}{75 \times 10^9 \text{ Pa} \times \frac{\pi}{32} (0.02)^4} = 0.12 \text{ rad} \Rightarrow \phi_B = 2.8\phi_C = 0.336 \text{ rad}$$

$$\phi_A = \phi_B + \frac{T_{AB} L_{AB}}{G I_{p,AB}} = 0.336 + \frac{56.10 \text{ N} \cdot \text{m} \times 0.6 \text{ m}}{75 \times 10^9 \text{ Pa} \times \frac{\pi}{32} (0.015)^4} = 0.426 \text{ rad}$$

6. A circular tube of inner radius $r_1 = 3 \text{ cm}$ and outer radius $r_2 = 4.5 \text{ cm}$ is subjected to a torque produced by forces P . The forces have their lines of action at a distance $b = 10 \text{ cm}$ outside of the tube. A torsional spring of stiffness $k_R = 100 \text{ kN} \cdot \text{m}/\text{rad}$ is added at the end of the tube to resist the twist. What is the maximum value of forces P if the allowable shear stress is $[\tau] = 100 \text{ MPa}$. Assume that the tube has a length of $L = 0.5 \text{ m}$ and shear modulus $G = 70 \text{ GPa}$.



Solution:

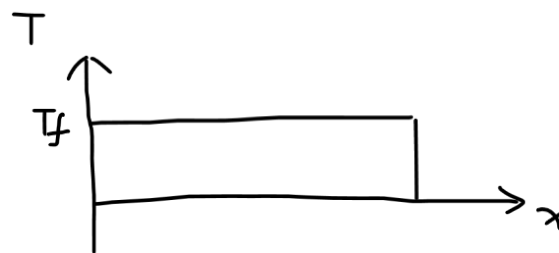
FBD:



Static equilibrium

$$T_f = T_p - T_R = 2P(r_2 + b) - T_R$$

TMD:



Using maximum stress to calculate maximum torque

$$I_p = \frac{\pi}{2}(r_2^4 - r_1^4) = 5.17 \times 10^{-6} \text{ m}^4$$

$$\tau_{max} = \frac{T_f r_2}{I_p} = [\tau] \Rightarrow T_f = \frac{[\tau] I_p}{r_2} = 11.49 \text{ kN} \cdot \text{m}$$

Twist angle at the end:

$$\phi = \frac{TL}{GI_p} = \frac{(2P(r_2 + b) - T_R)L}{GI_p}$$

Compatibility equation:

$$T_R = k_R \phi$$

Therefore

$$T_R = \frac{2P(r_2 + b)}{\left(\frac{GI_p}{k_R L} + 1\right)}$$

$$GI_p = 70 \times 10^9 \text{ Pa} \times 5.17 \times 10^{-6} \text{ m}^4 = 3.62 \times 10^5 \text{ N} \cdot \text{m}^2$$

Then

$$T_f = 2P(r_2 + b) - T_R = 2P(r_2 + b) - \frac{2P(r_2 + b)}{\left(\frac{GI_p}{k_R L} + 1\right)}$$

$$\frac{11.49 \text{ kN} \cdot \text{m}}{0.145 \text{ m}} = 2P \left(\frac{GI_p}{GI_p + k_R L} \right) = 2P \left(\frac{3.62 \times 10^5 \text{ N} \cdot \text{m}^2}{3.62 \times 10^5 \text{ N} \cdot \text{m}^2 + 50000 \text{ N} \cdot \text{m}^2} \right)$$

Solved

$$P = 45.1 \text{ kN}$$